

By TOUCH DEN · November 12, 2026

$ Installing Linux in a Virtual Machine

Video 1 · Why a VM, and Getting Ubuntu

Linux for Security · Section 2

In this video, you will learn:

- Why you install Linux in a VM, not on your real computer
- Which distribution to use
- How to get and check the Ubuntu image

Why a virtual machine

Everything in this course runs in a **VM** (How Computers Work, lesson 10), not on your real laptop.

- **Safety:** a mistake harms only the VM, never your real machine or data.
- **Snapshots:** restore a clean state in **seconds**, learning is fearless.
- **Clean:** no changes to your laptop; delete the VM and nothing remains.
- **The one rule:** the VM is **yours**, so "only test what you own" is met by construction.

Run it with a **hypervisor** (VirtualBox / UTM / VMware).

Which distribution

- **This course: Ubuntu LTS desktop**, friendly, used on real servers, well-documented.
- **Kali** (attacker's distro) comes **later**, in the lab and pentest courses. You don't need it now.

One Ubuntu VM.

Pick a hypervisor for your machine:

- **VirtualBox** (Windows/Linux/Intel Mac) · **UTM** (Apple Silicon) · **VMware** (personal use).

Get and check the image

Download the Ubuntu `.iso` (the installer) from the **official site only**.

Two security habits from the start:

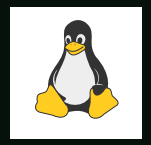
- **Official source only** (`ubuntu.com`), an OS from an untrusted site could be tampered with (supply-chain risk).
- **Check the checksum** (a hash, course 02): compute your file's hash and compare it to the official value. Match = genuine, unmodified.

Verify what you download, especially an operating system.

Recap: why a VM, and getting Ubuntu

- Install in a **VM**: safe, snapshot-able, isolated, and yours.
- Use **Ubuntu LTS**; Kali comes later.
- Download from the **official site** and **check the checksum**.

Next video: install Ubuntu, take a snapshot, and run the first update.



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Video 2 · Install, Snapshot, and Update

Linux for Security · Section 2

In this video, you will learn:

- Creating the VM and installing Ubuntu
- Snapshots: your safety net
- The first update

Create and install

1. **New VM** → point it at the Ubuntu `.iso`.
2. **Resources:** ~2 CPU cores, 4 GB RAM, 25 GB disk.
3. **Start** → it boots the installer (like booting from a USB stick).
4. In the installer: language, normal install, **use the whole (virtual) disk** (safe, it's the VM's disk).
5. **Account:** user **`dara`**, computer **`ubuntu`** → prompt reads `dara@ubuntu`.

Open **Terminal** → `dara@ubuntu:~$`. You have a real Linux machine.

Screenshot to capture

The installer account screen with name dara, computer ubuntu.

`assets/screenshots/03-linux/03-linux/ubuntu-create-account.png`

Snapshots: your safety net

A **snapshot** saves the VM's whole state so you can restore it instantly (course 02).

Right after install, take a snapshot named "clean install."

- **Mistakes become free:** break your VM → restore → clean in seconds.
- **Experiment boldly**, how you learn fastest.
- **A real practice:** analysts snapshot before detonating malware (course 29).

Snapshot before anything risky, and whenever the VM is known-good.

The first update

A fresh system is already behind on fixes. **Updating is a security task**, do it first on every new Linux system.

```
dara@ubuntu:~$ sudo apt update
dara@ubuntu:~$ sudo apt upgrade
```

- `sudo`: run with admin power (lesson 5); asks for your password.
- `apt update`: refresh the list of updates.
- `apt upgrade`: install them (`y` to confirm).

Then **snapshot again** ("updated"), a clean, patched starting point.

Recap: install, snapshot, update

- Create the VM, install Ubuntu with user `dara` on `ubuntu`, open the Terminal.
- **Snapshot** right away ("clean install"), restore in seconds, learn fearlessly.
- **Update first** (`sudo apt update && sudo apt upgrade`), then snapshot again.

Next lesson: start using it, the shell, the prompt, and the file system.